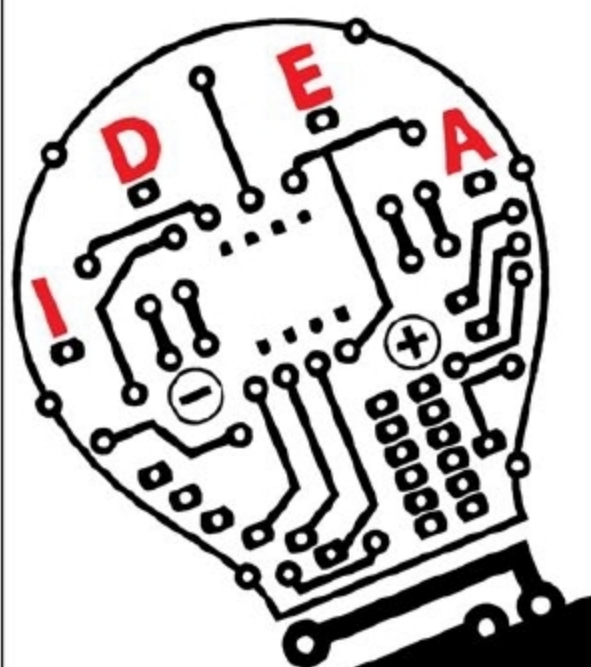


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## DO-IT-YOURSELF

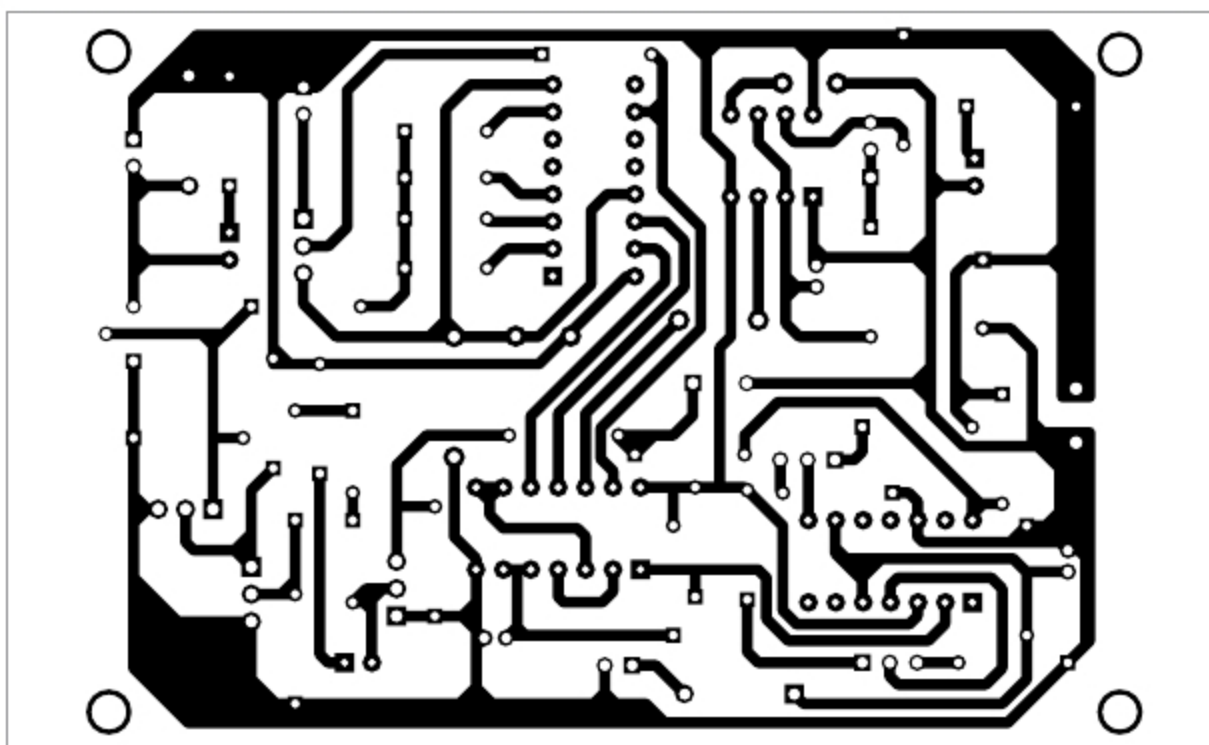


Fig. 3: Actual-size PCB layout of no-load overload protector for motorised circuit

### PARTS LIST

#### Semiconductors:

|               |                          |
|---------------|--------------------------|
| IC1           | - NE555 timer            |
| IC2           | - 4017 decade counter    |
| IC3           | - LM339 quad comparator  |
| IC4           | - CD4001 quad NOR gate   |
| T1, T3, T4    | - BC547 npn transistor   |
| T2            | - SL100 npn transistor   |
| ZD1           | - 5.1V Zener diode       |
| SCR1          | - BT169/XL08-40 scr      |
| D1-D5, D7, D8 | - 1N4148 signal diode    |
| D6            | - 1N4007 rectifier diode |
| LED1-LED3     | - 5mm LED                |

#### Resistors (all 1/4-watt, $\pm 5\%$ carbon):

|               |                              |
|---------------|------------------------------|
| R1, R9        | - 1.8-kilo-ohm               |
| R2            | - 100-kilo-ohm               |
| R3, R19, R20  | - 47-kilo-ohm                |
| R4, R16       | - 1-kilo-ohm                 |
| R5            | - 220-ohm                    |
| R6            | - 470-ohm                    |
| R7            | - 2.2-kilo-ohm               |
| R8, R10, R13, |                              |
| R18           | - 10-kilo-ohm                |
| R11, R12      | - 27-kilo-ohm                |
| R14           | - 15-kilo-ohm                |
| R15, R17, R21 | - 22-kilo-ohm                |
| VR1           | - 220-kilo-ohm potentiometer |
| VR2, VR3      | - 20-kilo-ohm potentiometer  |

#### Capacitors:

|    |                                  |
|----|----------------------------------|
| C1 | - 1000 $\mu$ F, 25V electrolytic |
| C2 | - 47 $\mu$ F, 25V electrolytic   |
| C3 | - 0.1 $\mu$ F ceramic disk       |
| C4 | - 0.01 $\mu$ F ceramic disk      |
| C5 | - 22 $\mu$ F, 25V electrolytic   |
| C6 | - 1 $\mu$ F, 25V electrolytic    |
| C7 | - 220 $\mu$ F, 25V electrolytic  |

#### Miscellaneous:

|            |                                 |
|------------|---------------------------------|
| CON1, CON2 | - 2-pin terminal connector      |
| S1         | - On/off switch                 |
| PZ1        | - Piezo buzzer                  |
| RL1        | - 12V, 2CO relay                |
| CT1        | - 10A, 50HZ current transformer |
|            | - 12V SMPS power supply/12V DC  |
|            | - 230AC operated motor for load |

from buzzer PZ1.

**Initial reset and extending power supply to load.** When power supply is initially switched on, 12V SMPS/12V DC extends DC supply voltage to the circuit. High output of counter may not be between Q0 and Q3. So, T1 and RL1 should not be energised until reset pin 15 of IC2 becomes high. In the beginning, no-load comparator A1 of IC3 gives low output, and output of gate N2 is high till voltage across C6 reaches an adequate level to change the output state of N2 to low.

Reset delay network comprises R12, C6 and N2. In the beginning of switch-on phase, input of gate N1 is high and low, and it gives low output. This is inverted by N3 to high, which resets IC2. Now, Q0 of IC2 becomes high and drives T1 and energises RL1.

### Construction and testing

An actual-size PCB layout of the no-load overload protector for the motorised circuit is shown in Fig. 3 and its components layout in Fig. 4.

Connect 12V SMPS or 12V DC to the circuit and enclose the PCB in a suitable PVC box. Since AC mains is used, use connectors with proper ratings and wires for AC input and output. Place the LEDs for power, normal output and overload in the front panel of PVC box. Make a hole on the box for the buzzer to indicate output tripped-off condition. Extend earth wire from AC mains to the load through the box.

To carry out no-load and overload